

ECE Probabilistic Systems Analysis

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Review

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad \text{assuming } P(B) > 0$$

- Multiplication rule:

$$P(A \cap B) = P(B) \cdot P(A | B) = P(A) \cdot P(B | A)$$

- Total probability theorem:

$$P(B) = P(A)P(B | A) + P(A^c)P(B | A^c)$$

- Bayes rule:

$$P(A_i | B) = \frac{P(A_i)P(B | A_i)}{P(B)}$$

Independence of Two Events

Definition of independence: $P(A \cap B) = P(A) \cdot P(B)$

- **“Defn:”** $P(B | A) = P(B)$
 - “occurrence of A provides no information about B ’s occurrence”
- Recall that $P(A \cap B) = P(A) \cdot P(B | A)$
- **Defn:** $P(A \cap B) = P(A) \cdot P(B)$
 - Symmetric with respect to A and B
 - applies even if $P(A) = 0$
 - implies $P(A | B) = P(A)$

A ball is selected from an urn containing two black balls, numbered 1 and 2, and two white balls, numbered 3 and 4. Let the events A , B , and C be defined as follows:

$A = \{(1, b), (2, b)\}$, “black ball selected”;

$B = \{(2, b), (4, w)\}$, “even-numbered ball selected”; and

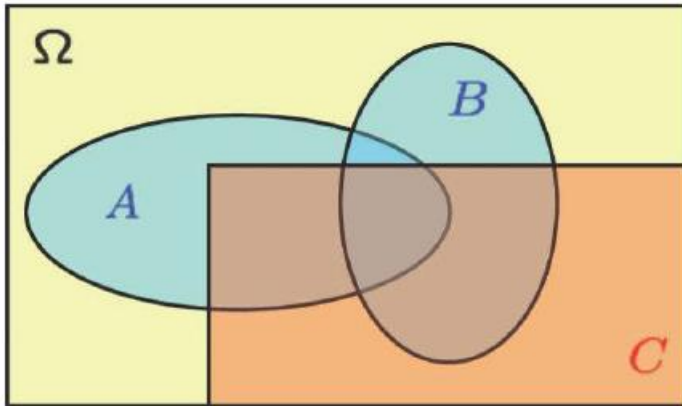
$C = \{(3, w), (4, w)\}$, “number of ball is greater than 2.”

Are events A and B independent? Are events A and C independent?

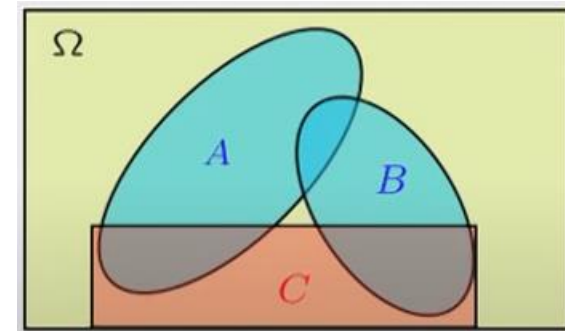
Concept	Meaning	Formulas
Disjoint	A and B cannot occur at the same time	$A \cap B = \emptyset,$ $P(A \cup B) = P(A) + P(B)$
Independent	A does not give any information about B	$P(A B) = P(A), P(B A) = P(B)$ $P(A \cap B) = P(A)P(B)$

Conditional Independence

- Conditional independence, given C , is defined as independence under the probability law $\mathbf{P}(\cdot | C)$



Assume A and B are independent



If we are told that C occurred, are A and B independent?

A box contains two coins: a regular coin and one fake two-headed coin ($P(H) = 1$). I choose a coin at random and toss it twice. Define the following events.

- A = First coin toss results in an H .
- B = Second coin toss results in an H .
- C = Coin 1 (regular) has been selected.

Find $P(A|C)$, $P(B|C)$, $P(A \cap B|C)$, $P(A)$, $P(B)$, and $P(A \cap B)$. Note that A and B are NOT independent, but they are *conditionally* independent given C .

Principles of Counting

4 shirts
3 ties
2 jackets

- r stages
- n_i choices at stage i

Number of choices is:

Principles of Counting

- Number of license plates with 3 letters and 4 digits =
- ... if repetition is prohibited =

Permutation

- Order of selection matters
- Without replacement

Number of ways of ordering n elements is:

Number of subsets of $\{1, \dots, n\}$ =

Example: Find the probability of a six roll die of a (six sided die) all give different number

- Number of outcomes that make the event happen:
- Number of elements in the sample space:

k-permutations

$$n(n-1)\cdots(n-k+1) = \frac{n(n-1)\cdots(n-k+1)(n-k)\cdots 2\cdot 1}{(n-k)\cdots 2\cdot 1}$$

Combination

- Order of selection is **NOT** matters
- Without replacement

$\binom{n}{k}$: number of k -element subsets
of a given n -element set

$$= \frac{n!}{k!(n-k)!}$$

constructing an **ordered** sequence of k **distinct** items:

Combination

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$\binom{n}{n} =$$

$$\binom{n}{0} =$$

$$\sum_{k=0}^n \binom{n}{k} =$$

Binomial probabilities

Example:

You have n independent coin tosses. Suppose $P(H) = p$, find the probability that head shows K times with all possible sequences.

sequence of independent **Bernoulli trials**.

- $P(HTTHHH) =$
- $P(\text{particular sequence}) =$
- $P(\text{particular } k\text{-head sequence})$

$$P(k \text{ heads}) =$$

Coin tossing problem

- event B : 3 out of 10 tosses were “heads”.

- Given that B occurred,
what is the (conditional) probability
that the first 2 tosses were heads?

Partitions

Ten people have a potluck. Five people will be selected to bring a main dish, three people will bring drinks, and two people will bring dessert. How many ways they can be divided into these three groups?

I roll a die 18 times. What is the probability that each number appears exactly 3 times.

Example 1.30. A class consisting of 4 graduate and 12 undergraduate students is randomly divided into four groups of 4. What is the probability that each group includes a graduate student?

Summary of Counting Results

- Permutations of n objects: $n!$
- k -permutations of n objects: $n!/(n - k)!$
- Combinations of k out of n objects: $\binom{n}{k} = \frac{n!}{k!(n - k)!}$
- Partitions of n objects into r groups with the i th group having n_i objects:

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1! n_2! \cdots n_r!}.$$

- (a) A coin is tossed twice and the sequence of heads and tails is noted. Let S be the sample space of this experiment. Find all subsets of S .
- (b) A coin is tossed twice and the number of heads is noted. Let S' be the sample space of this experiment. Find all subsets of S' .

A die is tossed and the number of dots facing up is noted.

- (a) Find the probability of the elementary events under the assumption that all faces of the die are equally likely to be facing up after a toss.
- (b) Find the probability of the events: $A = \{\text{more than 3 dots}\}$; $B = \{\text{odd number of dots}\}$.
- (c) Find the probability of $A \cup B$, $A \cap B$, A^c .

2.77. A nonsymmetric binary communications channel is shown in Fig. P2.3. Assume the input is “0” with probability p and “1” with probability $1 - p$.

- (a)** Find the probability that the output is 0.
- (b)** Find the probability that the input was 0 given that the output is 1. Find the probability that the input is 1 given that the output is 1. Which input is more probable?

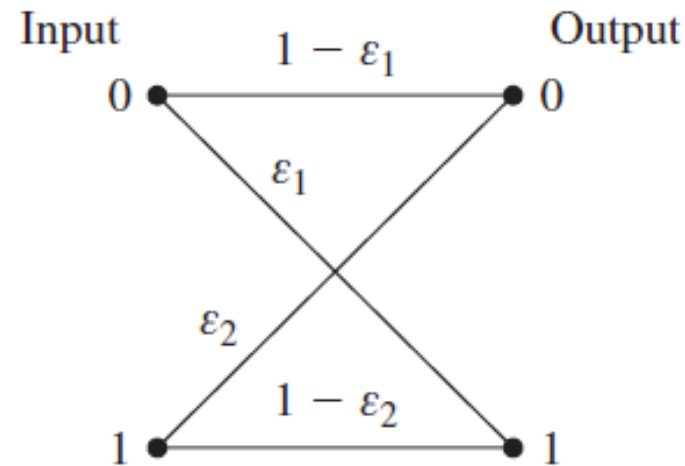


FIGURE P2.3

Q2/ A die is tossed twice and the number of dots facing up in each toss is counted and noted in the order of occurrence.

1- Find the sample space.

2- Find the set A corresponding to the event “number of dots in first toss is bigger than number of dots in second toss.”

3- Find the set B corresponding to the event “number of dots in first toss is 6.”

4- Does A subset of B or does B subset of A ?

5- Find $A \cap B^C$.
